

■ Hospital OPD Analyzer

Weekly Performance & Overcrowding Report

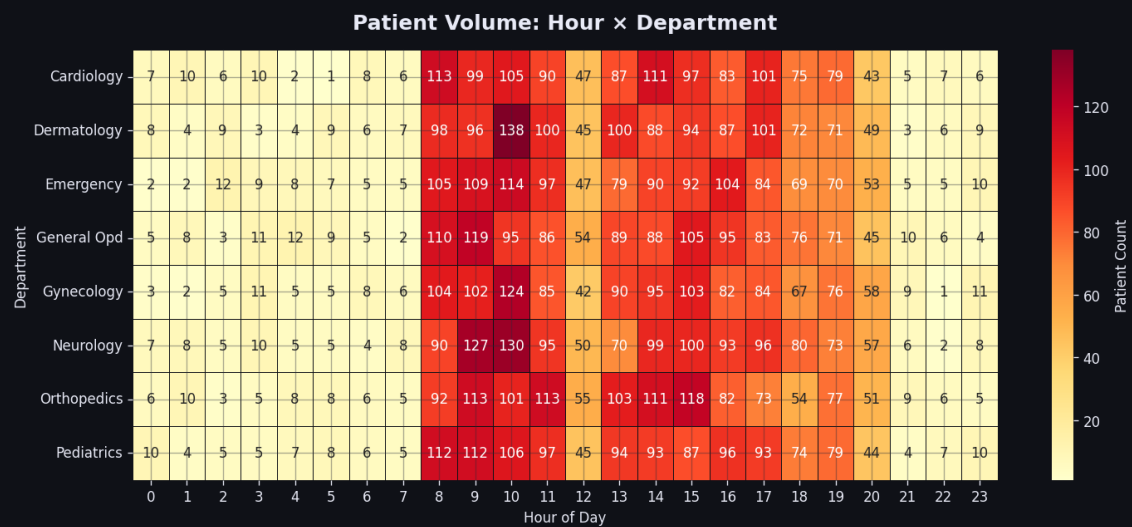
Data Period: 2024-01-01 00:00:00 → 2024-03-30 00:00:00

Generated: 22 April 2026, 04:25 AM

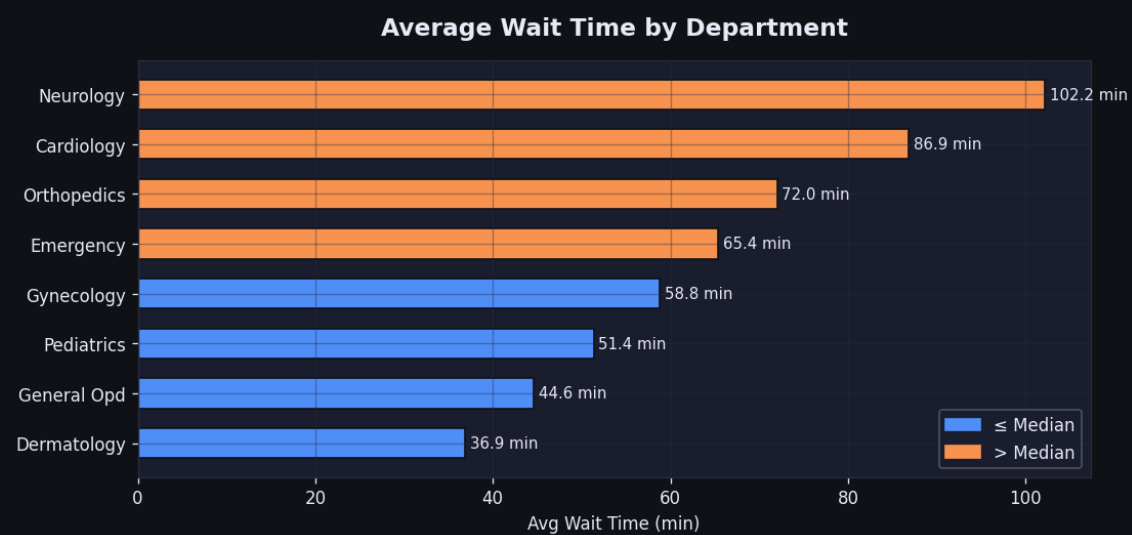
Total Records	Departments	Date Range
9602	8	2024-01-01 00:00:00 – 2024-03-30 00:00:00

Visualizations

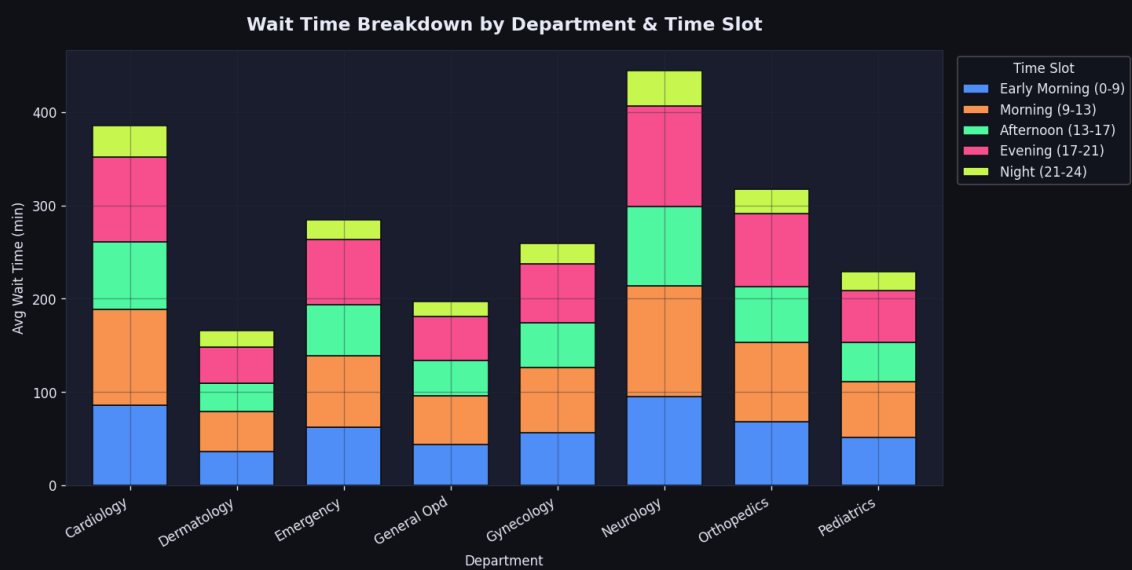
Patient Volume Heatmap (Hour × Department)



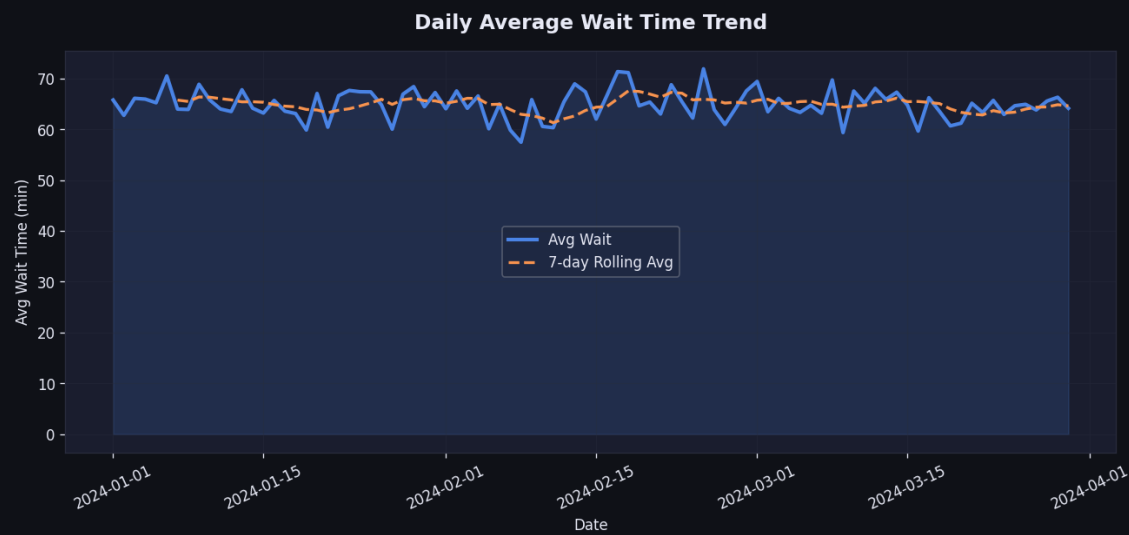
Average Wait Time by Department



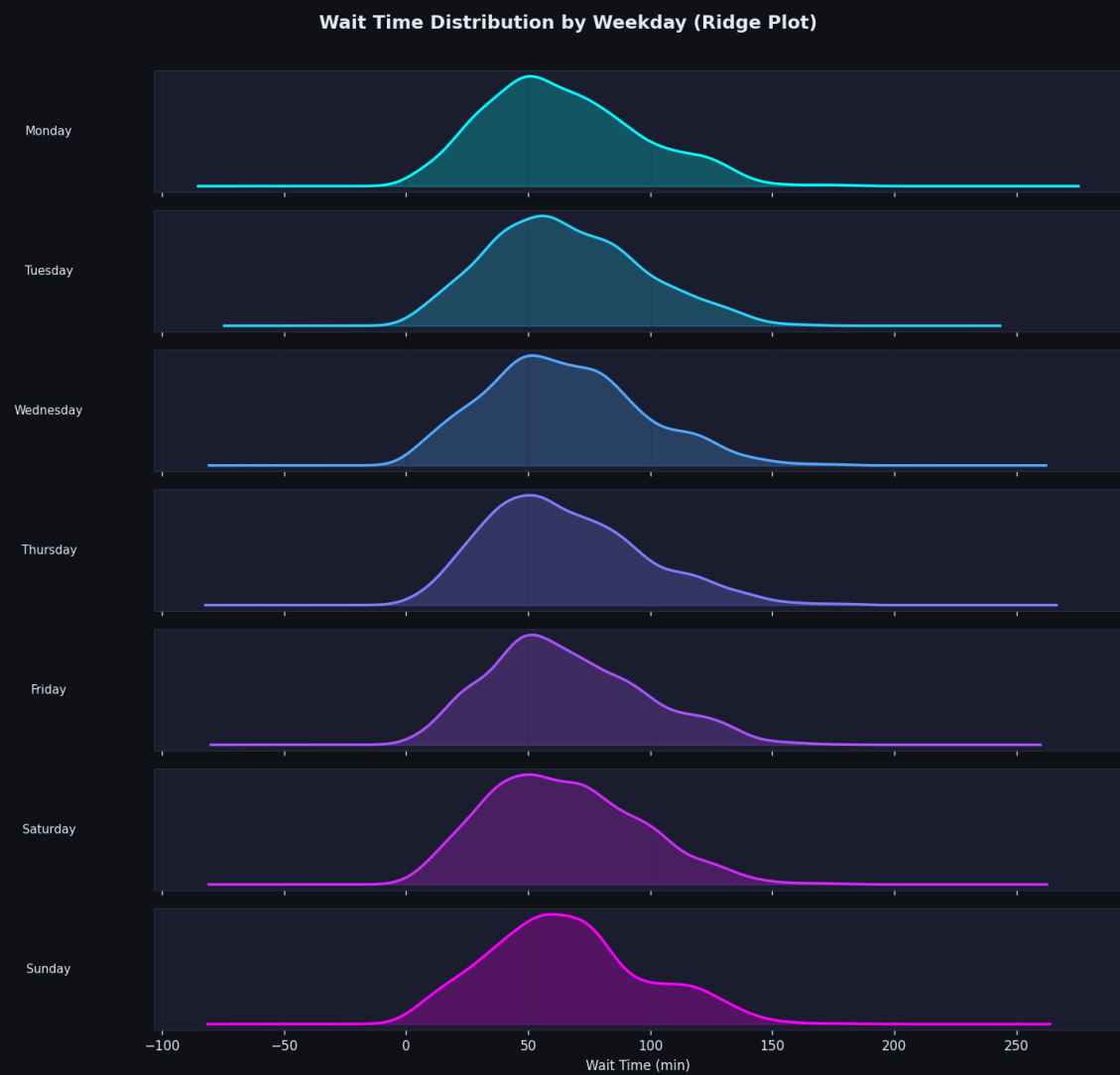
Wait Time Breakdown by Time Slot



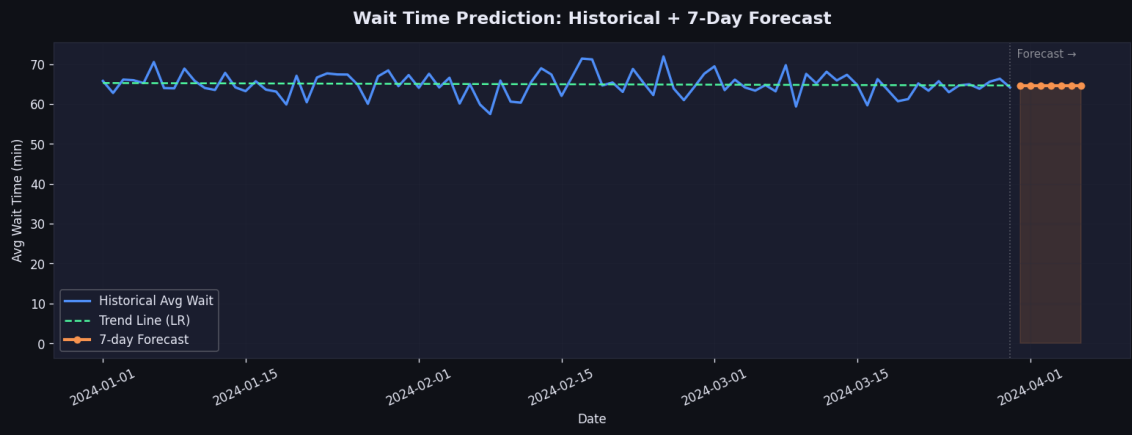
Daily Wait Time Trend



Weekday Wait Distribution (Ridge Plot)



7-Day Wait Time Prediction



■ Key Insights & Recommendations

■ *Neurology has the highest average wait*

→ Average wait time is 102 min. Recommend adding 1–2 extra staff during peak hours.

■ *Neurology causes max cascading delay*

→ On stressed days, wait time jumps by ~18 min extra. Doctor cover plan needed.

■ *50 extreme delay events detected*

→ Extreme delays ($>2.5\sigma$ above dept mean) suggest occasional system failures or data entry errors. Audit required.

■ *Neurology: peak at 8:00*

→ Avg wait reaches 129.0 min. Schedule extra consultants between 7:00–10:00.

■ *Cardiology: peak at 11:00*

→ Avg wait reaches 113.9 min. Schedule extra consultants between 10:00–13:00.

■ *Orthopedics: peak at 9:00*

→ Avg wait reaches 95.0 min. Schedule extra consultants between 8:00–11:00.

■ *Friday is the busiest day*

→ ~1599 patients on average. Consider full staffing + floating doctors on this day.

■ 7-Day Wait Time Forecast

Date	Day	Predicted Wait (min)
2024-03-31	Sunday	64.6
2024-04-01	Monday	64.6
2024-04-02	Tuesday	64.6
2024-04-03	Wednesday	64.6
2024-04-04	Thursday	64.6
2024-04-05	Friday	64.6
2024-04-06	Saturday	64.6

Model Performance → MAE: 2.24 min | R²: 0.004 | Trend: ■ Stable

■ Doctor Absence Impact Analysis

Most impacted department: **Neurology** (+18.4 min cascading delay)

Department	Normal Avg	Stressed Avg	Cascading Delay	Stressed Days
Neurology	100.5 min	118.9 min	+18.4 min	9
Cardiology	85.6 min	99.4 min	+13.8 min	9
Orthopedics	70.8 min	84.2 min	+13.5 min	9
Emergency	64.1 min	77.4 min	+13.3 min	9
Gynecology	57.9 min	69.4 min	+11.5 min	9
Pediatrics	49.7 min	61.3 min	+11.5 min	9
General Opd	43.5 min	54.8 min	+11.3 min	9
Dermatology	36.1 min	44.9 min	+8.8 min	9